

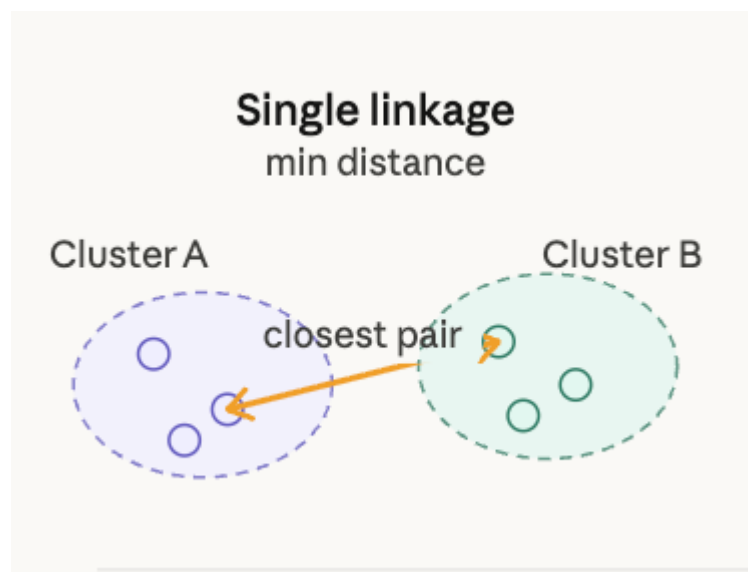
Linkage

Linkage calculates the distance between clusters before they are merged or divided. There are different types of linkage, each of which has its own purpose.

1. Single Linkage

It uses the minimum distance between any two points, one from each cluster. It finds the closest pair across the two groups.

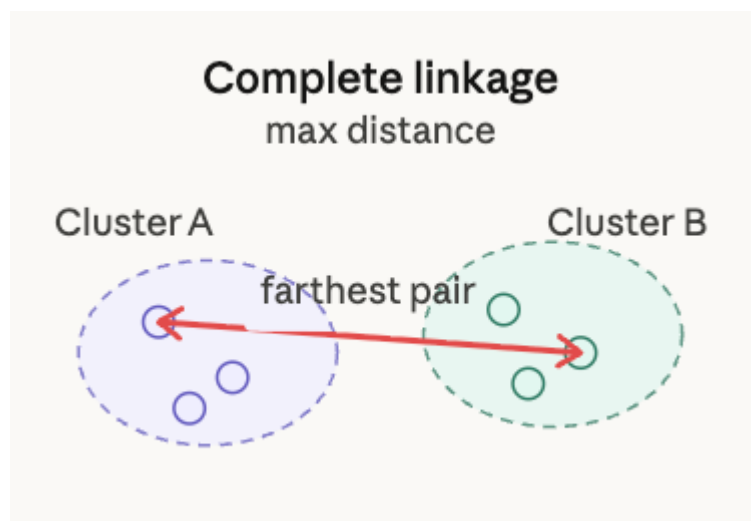
$$d(A, B) = \min_{x \in A, y \in B} d(x, y)$$



2. Complete Linkage

It uses the maximum distance between any two points across the clusters. It finds the farthest pair

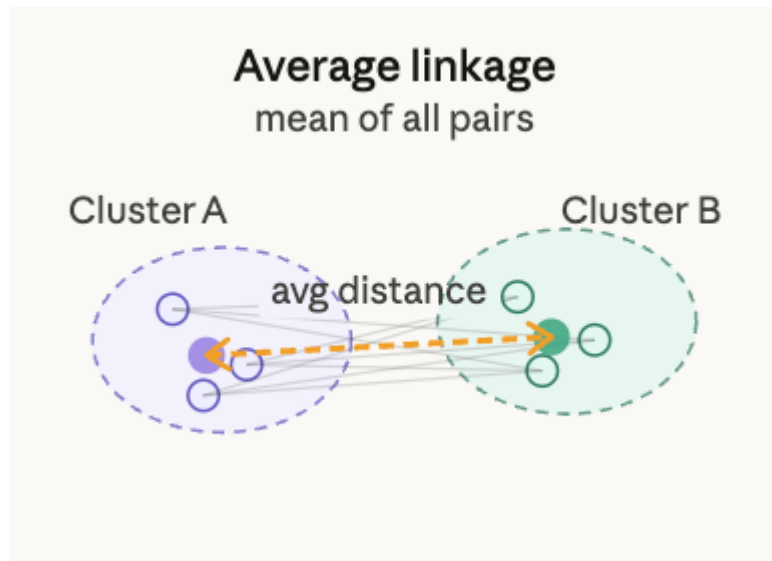
$$d(A, B) = \max_{x \in A, y \in B} d(x, y)$$



3. Average Linkage

It computes the average distance between all pairs of points across the two clusters.

$$d(A, B) = \frac{1}{|A| \cdot |B|} \sum_{x \in A} \sum_{y \in B} d(x, y)$$

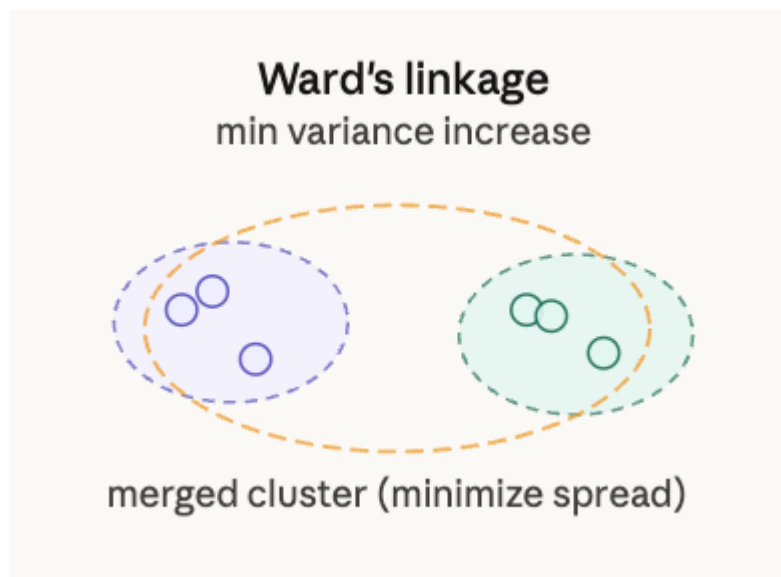


4. Ward's Linkage

It merges the two clusters that result in the smallest increase in total within-cluster variance.

It is less intuitive but often produces the most compact, balanced clusters.

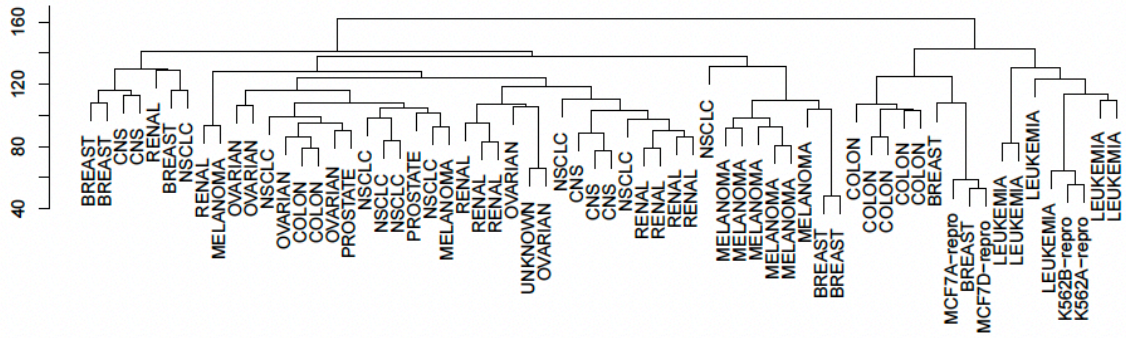
$$d(A, B) = \sqrt{\frac{2|A||B|}{|A| + |B|}} \cdot \|\bar{x}_A - \bar{x}_B\|$$



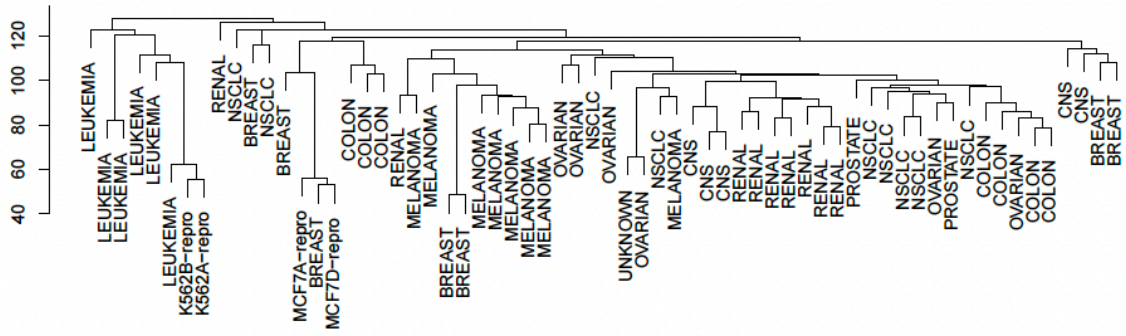
Where:

- (A, B) = two clusters being compared
- x, y = individual data points in cluster A and B
- $d(x, y)$ = Distance between point x and y
- $|A|, |B|$ = Numbers of points in cluster A and B
- $\bar{x}_A, \bar{x}_B$ = Centroid (mean) of cluster A and B
- $\|\cdot\|$ = Euclidean Distance between two centroid

Complete Linkage



Average Linkage



Single Linkage

